

MODELING & EVALUATION REPORT

Beyond the Canopy: Predicting Deforestation Risk Along Brazil's BR-163 Highway



Section 1: Data Preparation for Training

This section describes how the raw dataset was transformed into a form suitable for machine learning. Each step is explained and justified in terms of its effect on model quality.

1a. Feature Engineering

Encoding Categorical Variables

The dataset contains one categorical variable: `land_cover`, which records the land use type of each grid cell (e.g., forest, pasture, mining, agriculture). This variable is nominal, meaning there is no meaningful numeric order between its categories. Encoding it with label encoding would falsely imply a numeric ordering between categories such as forest, pasture, and mining. One-hot encoding was used instead, converting each land cover category into its own binary 0/1 column. This allows every model to treat each land cover type as a completely independent signal with no assumed ordering.

Redundant Features Removed Based on EDA

Four features were dropped before modelling based on the correlation analysis in the Data Analysis Report:

- `evi` was removed because it has a correlation of $r = 0.995$ with `ndvi_mean`. These two vegetation indices are nearly perfectly collinear and carry identical information. Keeping both adds dimensionality without adding any new signal.
- `fire_count_buffer` was removed because it correlates at $r = 0.714$ with `fire_count_cell`, and the log-transformed version already captures the neighbourhood fire signal in a better-distributed form.
- `dist_highway_km` was replaced by `log_dist_highway`. The raw distance is heavily right-skewed and the log transform compresses the long tail into a more informative feature.
- `dist_road_km` was replaced by `log_dist_road` for the same reason as above.

New Features

The four log transforms (`log_dist_highway`, `log_dist_road`, `log_fire_cell`, `log_fire_buffer`) were created in the Data Analysis Report and carried forward unchanged. No additional new features were created in this deliverable.

Handling Class Imbalance

The training set contains 223,657 intact cells and 50,598 deforested cells, a ratio of approximately 4.42 to 1. Rather than resampling with SMOTE or oversampling, all classifiers were trained with `class_weight='balanced'`, which automatically assigns a higher loss penalty to minority class misclassifications. Sklearn computes a weight of 0.61 for intact cells and 2.71 for deforested cells, meaning every misclassified deforested cell is penalised 4.4 times more heavily than an intact cell.

SMOTE was not used because it is designed for severe imbalance (ratios of 95/5 or worse) where too few real minority examples exist. Here there are 50,598 real deforested cells in the training set, far more than enough to learn from without generating synthetic ones. The final XGBoost model achieved a recall of 0.947 on the deforested class, confirming `class_weight='balanced'` was sufficient.

1b. Data Splitting Strategy

An 80/20 stratified train-test split was used. The dataset contains 342,819 rows, producing a training set of approximately 274,255 rows and a test set of approximately 68,564 rows. Stratified splitting ensures both sets preserve the same 81.6% / 18.4% class ratio as the full dataset. Without stratification, random chance could produce an unrepresentative test set.

`random_state=42` was set throughout to ensure full reproducibility: anyone running this notebook on the same data will obtain identical splits. The test set was set aside immediately after splitting and was not touched during model training, cross-validation, or hyperparameter tuning. It was used exclusively for the final evaluation in Sections 3 and 4.

Spatial Autocorrelation Note

The ideal approach for geospatial data is geographic block splitting, where a contiguous spatial region is held out as the test set. Because a spatial block index is not available in this dataset, stratified random splitting was used and this is acknowledged as a limitation. Spatial block cross-validation remains the gold standard for this problem type.

1c. Feature Scaling

`StandardScaler` (zero mean, unit variance) was applied to all continuous features, but only for Logistic Regression. It was not applied to Random Forest or XGBoost. Logistic Regression uses gradient-based optimisation and is sensitive to feature scale: without scaling, features measured in large units dominate the gradient updates while small-scale features are effectively ignored. Tree-based models split on thresholds rather than magnitudes and are completely scale-invariant, so scaling them would waste computation with no effect on results.

Data Leakage Prevention

The scaler was fit exclusively on `X_train` and then applied to `X_test` using `transform` only, never `fit_transform`. Fitting the scaler on the full dataset before splitting would leak test set distribution statistics into the training process, artificially inflating performance estimates. This was avoided.

A Note on False Negatives

Throughout this project, minimising false negatives is treated as the primary operational concern. A false negative means a grid cell that is actively being deforested is predicted as intact. In practice, this means the cell receives no monitoring attention and no enforcement response while clearing continues. For BR-163 conservation work, this is a far more costly error than a false alarm. This priority directly shapes metric choice and final model selection in all sections that follow.

Section 2: Baseline Models

Before building and tuning more complex models, it is important to establish a performance floor. A baseline answers a fundamental question: is the model actually learning anything useful, or could a much simpler rule achieve similar results? Every metric reported in later sections should be interpreted relative to what a naive predictor can already achieve.

Four models are evaluated here using default hyperparameters with no tuning. This serves two purposes: it gives a reference floor to measure how much improvement comes from tuning in Section 3, and it lets us compare all model families on equal footing before any optimisation is applied. All real models use `class_weight='balanced'` to account for the 4.42:1 class imbalance. A dummy majority-class classifier is also included to demonstrate concretely why accuracy alone is a misleading metric on this dataset.

Evaluation Metrics: Choice and Justification

Choosing the right metrics is just as important as choosing the right model. For an imbalanced conservation problem like this one, the standard metric of accuracy is actively misleading. The metrics used throughout this report are selected because they each reveal something different and meaningful about model behaviour.

Primary Metric: AUC-ROC. AUC-ROC (Area Under the Receiver Operating Characteristic Curve) measures how well the model separates deforested cells from intact ones across every possible classification threshold simultaneously. A score of 0.5 means the model is no better than random guessing; a score of 1.0 means perfect separation. AUC-ROC is chosen as the primary metric because it is completely unaffected by class imbalance. Whether the dataset is 80/20 or 99/1, AUC-ROC evaluates the model's fundamental discriminative ability independent of any threshold decision.

Secondary Metric: Recall on the Deforested Class. Recall measures what fraction of all actually deforested cells the model successfully identifies. A False Negative in this problem means a grid cell that is actively being cleared is predicted as intact. In operational terms, this cell receives no monitoring attention and no enforcement response while deforestation continues. This is the most costly possible error. Maximising recall directly minimises the number of real clearing events the model lets slip through undetected, which is why it is treated as the second most important metric throughout this report.

Supporting Metric: F1-Score on the Deforested Class. F1 is the harmonic mean of Precision and Recall. It rewards models that are both good at catching deforested cells (high recall) and not excessively over-flagging intact cells (high precision). It is reported alongside recall to give a complete picture of minority class performance.

Why Accuracy Is Not Used for Model Selection. A classifier that predicts every single cell as Intact scores 81.6% accuracy on this dataset while catching zero deforested cells. Accuracy is dominated by the majority class and gives no information about how the model handles the class that actually matters. It is reported for completeness only.

Metric	Role	Why It Is Appropriate Here
AUC-ROC	Primary	Threshold-independent, robust to class imbalance
Recall	Secondary	Directly measures how many deforested cells are caught

F1-Score	Supporting	Balances recall against over-flagging intact cells
Accuracy	Reported only	Misleading on imbalanced data, not used for selection

Why class_weight='balanced' Instead of SMOTE

Without any correction, a model will naturally bias toward predicting Intact on ambiguous cases because that is where the majority of training examples come from. class_weight='balanced' addresses this by assigning a higher loss penalty to minority class errors during training. The weights are computed automatically: 0.61 for Intact and 2.71 for Deforested. For XGBoost the equivalent parameter is scale_pos_weight = 4.42.

SMOTE was not used because it is designed for severe imbalance scenarios (ratios of 95/5 or worse) where the model sees too few real minority class examples to learn the pattern from real data alone. This dataset contains 50,598 real deforested cells in the training set. Generating synthetic interpolated samples would introduce noise without adding genuine information. The 81/19 ratio is moderate and class_weight='balanced' is the standard and sufficient correction at this level.

Baseline Results: Interpretation

Majority Class classifier: Scores 81.6% accuracy while catching zero deforested cells (F1 = 0.000, AUC = 0.500). This result alone justifies why accuracy is excluded from model selection: an entirely useless predictor scores above what most would consider bad on this metric.

Logistic Regression: Performs surprisingly well at default settings, reaching AUC = 0.981 and F1 = 0.844. The EDA showed strong linear signals in log fire count, NDVI change, and log distance to road, meaning a linear decision boundary can already capture most of the separation.

Decision Tree: Has the lowest AUC among real models at 0.891. Default decision trees grow to unlimited depth and tend to memorise training data rather than generalise. The train-test AUC gap of 0.109 confirms this overfitting behaviour.

Random Forest: Leads on F1 at 0.874 and reaches AUC = 0.988. Averaging predictions across 100 trees cancels out the individual tree overfitting, bringing the train-test gap down to just 0.012.

XGBoost: Leads on AUC at 0.989 at default settings, with a train-test gap of only 0.005. Sequential boosting is more conservative by nature: each tree corrects only the residual errors of the previous one rather than fitting the full signal, which naturally limits overfitting.

Section 3: Model Training, Evaluation and Justification

Three model types are trained and evaluated: Logistic Regression, Random Forest, and XGBoost. These three were chosen because they represent fundamentally different learning approaches. A linear model, a parallel ensemble of trees, and a sequential boosting ensemble allow for a meaningful comparison of how different model families handle the same problem. The goal is not to try everything but to justify each choice and interpret every result in the context of the BR-163 conservation problem.

For every model, the same evaluation procedure applies: 5-fold stratified cross-validation on the training set for model selection and hyperparameter tuning, followed by a single final evaluation on the held-out test set. The test set was never used during training or tuning at any stage.

3.1 Logistic Regression

a) Model Justification

Logistic Regression works by modelling the log-odds of deforestation as a weighted linear combination of input features, then passing that through a sigmoid function to produce a probability between 0 and 1. The model assumes that a linear decision boundary can meaningfully separate deforested from intact cells in feature space.

The EDA supports this assumption. Log fire count, NDVI change, and log distance to road all show strong, roughly monotonic relationships with deforestation probability, meaning that as each of these values changes, deforestation risk moves consistently in one direction. A linear boundary is capable of capturing the majority of this signal without needing to model complex non-linear interactions. The model is also highly interpretable: each feature coefficient directly represents that feature's contribution to the log-odds of deforestation, making it easy to explain predictions to non-technical stakeholders. The model makes no assumption about the distribution of input features; it only requires that the relationship between each feature and the log-odds of the target is approximately linear, which the EDA confirms.

b) Training Approach

Features were standardised using StandardScaler before training, as described in Section 1c. This is required for Logistic Regression because gradient-based optimisation is sensitive to feature scale. Without scaling, features measured in large units dominate the gradient updates and features measured in small units are effectively ignored.

The only hyperparameter tuned was C, the inverse of regularisation strength. A small value of C applies strong regularisation by shrinking coefficients toward zero. A large value of C reduces regularisation, allowing the model to fit more complex boundaries. A log-scale grid search was conducted over five values: 0.01, 0.1, 1, 10, and 100.

Tuning was performed using 5-fold stratified cross-validation on the training set only, with AUC-ROC as the scoring metric. Stratified folds preserve the 4.42:1 class ratio across every fold. The test set was not used at any point during this process. Reproducibility was ensured by setting `random_state=42` on both the cross-validation splitter and the model itself.

c) Evaluation Metrics and Results Interpretation

The model was evaluated on the held-out test set of 68,564 rows after tuning was complete.

Best C = 0.01. The model strongly prefers heavy regularisation. With over 274,000 training rows and only 27 features, the model is nowhere near the risk of overfitting. Strong regularisation stabilises the coefficients and the large training volume means the true signal is learned reliably even with a constrained model.

AUC-ROC = 0.981. Excellent for a purely linear model. The model correctly ranks the deforested cell as higher risk 98.1% of the time across all thresholds simultaneously.

Recall = 0.924. The model correctly identifies 11,686 out of 12,649 genuinely deforested cells. This is the metric that matters most in a conservation context: a high recall means fewer real clearing events slip through undetected.

F1 = 0.844. Reflects a reasonable balance between catching deforested cells and not over-flagging intact ones.

Confusion Matrix Interpretation:

- 963 False Negatives: deforested cells predicted as intact. Each represents a real clearing event that would receive no monitoring or enforcement response. This is the most costly error type.
- 3,365 False Positives: intact cells wrongly flagged as deforested. A less severe error operationally: it means some monitoring resources are directed toward cells that do not need attention, but no real deforestation goes undetected.

Improvement from tuning: Tuning produced negligible improvement. AUC remained at 0.981. Logistic Regression was already operating near its performance ceiling. The limiting factor is the linear decision boundary itself, not the regularisation parameter. Non-linear interactions between fire activity, road proximity, and land cover cannot be captured by this model family regardless of C. This directly motivates moving to tree-based ensemble models.

3.2 Random Forest

a) Model Justification

Random Forest builds a large number of decision trees, each trained on a random subset of training rows and a random subset of features, then combines their predictions by majority vote. This ensemble approach directly addresses the core weakness of the single Decision Tree in the baseline: an individual tree grown to unlimited depth achieved a train AUC of 1.000 but a test AUC of only 0.891, a gap of 0.109 reflecting severe overfitting. Averaging across many trees cancels out the individual noise each tree memorises.

The EDA revealed that deforestation risk is driven by the interaction of multiple factors simultaneously. Road proximity, fire activity, land cover type, and vegetation decline combine in non-linear ways that a linear model cannot capture. Random Forest handles these interactions naturally by splitting on combinations of features at different depths, without requiring manual feature transformation. It is also non-parametric with no distributional assumptions, robust to the moderate remaining correlation between fire-related features, and natively handles the mix of continuous and binary one-hot encoded features.

b) Training Approach

No feature scaling was applied. Tree-based models split on feature thresholds rather than magnitudes, so feature scale has no effect on model performance. Three hyperparameters were tuned:

- `n_estimators`: number of trees in the forest. Values of 200 and 500 were searched. More trees generally produce more stable predictions with diminishing returns beyond a threshold.
- `max_depth`: limits individual tree depth. Values of 10, 20, and None (unlimited) were searched. Unlimited depth is appropriate here because averaging across many trees regularises the individual overfitting that a single unlimited-depth tree suffers from.

- `min_samples_leaf`: minimum samples required at a leaf node. Values of 1 and 5 were searched. Higher values prevent the model from fitting splits that apply to only a handful of rows.

Tuning was performed using 5-fold stratified cross-validation on the training set only, with AUC-ROC as the scoring metric. Stratified folds preserve the 4.42:1 class ratio across every fold. `random_state=42` was set for reproducibility. `n_jobs=-1` parallelised training across all available CPU cores.

c) Evaluation Metrics and Results Interpretation

The model was evaluated on the held-out test set of 68,564 rows after tuning was complete.

Best parameters: `max_depth=None`, `min_samples_leaf=1`, `n_estimators=500`. Full-depth trees with 500 estimators. Unlimited depth works here because averaging 500 trees suppresses the noise any individual tree memorises.

AUC-ROC = 0.989. Second highest AUC of all models and an improvement over Logistic Regression (0.981). Random Forest captures non-linear signal that a linear boundary misses.

Recall = 0.855. Correctly identifies 10,816 out of 12,649 deforested cells. This is notably lower than Logistic Regression's recall of 0.924. Random Forest is more conservative: it only flags a cell as deforested when multiple trees agree with high confidence.

F1 = 0.876. Highest F1 of all models, reflecting Random Forest's strong precision. It is the most efficient model at minimising false alarms, but this comes at the cost of higher false negatives.

Confusion Matrix Interpretation:

- 1,833 False Negatives: almost double the false negative count of Logistic Regression (963) and nearly three times that of XGBoost (676). Despite its high F1 and AUC, Random Forest's conservative behaviour makes it less suitable for this conservation problem where catching deforestation is the priority.
- 1,231 False Positives: the lowest false positive count of all models, confirming Random Forest is precise but not aggressive enough in catching deforestation.

Improvement from tuning: AUC improved from 0.988 to 0.989 and F1 from 0.874 to 0.876. Modest but consistent gains confirm the optimal configuration uses 500 full-depth trees.

3.3 XGBoost

a) Model Justification

XGBoost is a gradient boosting algorithm. Unlike Random Forest, which builds hundreds of trees independently in parallel and averages their predictions, XGBoost builds trees sequentially: each new tree is trained specifically to correct the residual errors that all previous trees still get wrong. Every additional tree is focused precisely on the hardest cases the model currently misclassifies. The result is a model that improves iteratively and converges to a highly precise decision boundary.

This sequential correction mechanism makes XGBoost particularly well suited to the BR-163 deforestation problem. The EDA identified a non-linear, multi-factor risk space where fire activity, road proximity, land cover type, and vegetation decline interact in ways no single feature can capture alone. Gradient boosting excels at learning exactly these kinds of complex, high-order interactions across tabular features, which is reflected in XGBoost being the strongest baseline model at default settings with an AUC of 0.989.

XGBoost makes no distributional assumptions, does not require feature scaling, handles the mix of continuous and binary one-hot encoded columns natively, and manages class imbalance through

scale_pos_weight = 4.42, the ratio of intact to deforested cells in the training set. Misclassifying a deforested cell is penalised 4.42 times more heavily than misclassifying an intact cell during training.

b) Training Approach

No feature scaling was applied. XGBoost is a tree-based model that splits on thresholds and is completely scale-invariant. Four hyperparameters were tuned:

- `n_estimators`: number of sequential boosting rounds. Values of 200 and 500 were searched.
- `max_depth`: depth of each individual tree. Unlike Random Forest where unlimited depth is offset by averaging, XGBoost trees need to be kept relatively shallow because each one directly influences the next. Values of 3 and 6 were searched.
- `learning_rate`: how much each new tree contributes to the final prediction. A lower rate means more conservative and generalisable models. Values of 0.01 and 0.1 were searched.
- `subsample` was fixed at 0.8, meaning each tree trains on a random 80% of training rows. This introduces randomness similar to Random Forest's row sampling and reduces overfitting.

Tuning was performed using 5-fold stratified cross-validation on the training set only, with AUC-ROC as the scoring metric. The test set was not accessed at any point during tuning. `random_state=42` was set for reproducibility and `n_jobs=-1` parallelised the grid search.

c) Evaluation Metrics and Results Interpretation

The model was evaluated on the held-out test set of 68,564 rows after tuning was complete.

Best parameters: `learning_rate=0.1`, `max_depth=6`, `n_estimators=500`. A moderate learning rate with 500 boosting rounds and depth-6 trees. The preference for depth-6 over depth-3 indicates that the deforestation risk boundary is genuinely complex and requires trees capable of capturing multi-factor interactions.

AUC-ROC = 0.990. Highest AUC of all models. XGBoost correctly ranks deforested cells above intact ones 99.0% of the time across all classification thresholds.

Recall = 0.947. Highest recall of all models. The model correctly identifies 11,973 out of 12,649 genuinely deforested cells. In a conservation context, recall directly measures how many real clearing events the model catches before they go unmonitored. XGBoost catches 1,157 more deforested cells than Random Forest and 287 more than Logistic Regression.

F1 = 0.871. Slightly below Random Forest's F1 of 0.876, reflecting the trade-off XGBoost makes: it catches more deforested cells (higher recall) but flags more intact cells as well (lower precision). For this conservation problem, this is the correct trade-off to make.

Confusion Matrix Interpretation:

- 676 False Negatives: by far the lowest false negative count of all models. Logistic Regression produced 963, Random Forest produced 1,833, and Decision Tree produced 2,285. Reducing false negatives from 1,833 to 676 means 1,157 additional deforestation events are caught and flagged for action. In a conservation setting this difference is operationally significant.
- 2,879 False Positives: higher than Random Forest's 1,231, which is the direct cost of XGBoost's more aggressive minority class detection. However, directing monitoring resources toward a cell that turns out to be intact is far less severe than missing a cell that is actively being cleared.

Improvement from tuning: AUC improved from 0.989 at default settings to 0.990 after tuning, and false negatives dropped significantly due to the optimised combination of learning rate and tree depth. XGBoost is selected as the final model: it achieves the highest AUC, the highest recall, and the lowest false negative count of all models evaluated.

Section 4: Model Comparison

All models are compared across multiple metrics on the same held-out test set of 68,564 rows, which was never seen during training or tuning. The majority-class baseline is included to show the performance floor.

Model	AUC-ROC	F1	Recall	Precision	False Neg.	False Pos.
Majority Class	0.500	0.000	0.000	0.000	12,649	N/A
Logistic Reg.	0.981	0.844	0.924	0.765	963	3,365
Decision Tree	0.891	0.825	0.819	0.831	2,285	2,476
Random Forest	0.989	0.876	0.855	0.898	1,833	1,231
XGBoost	0.990	0.871	0.947	0.813	676	2,879

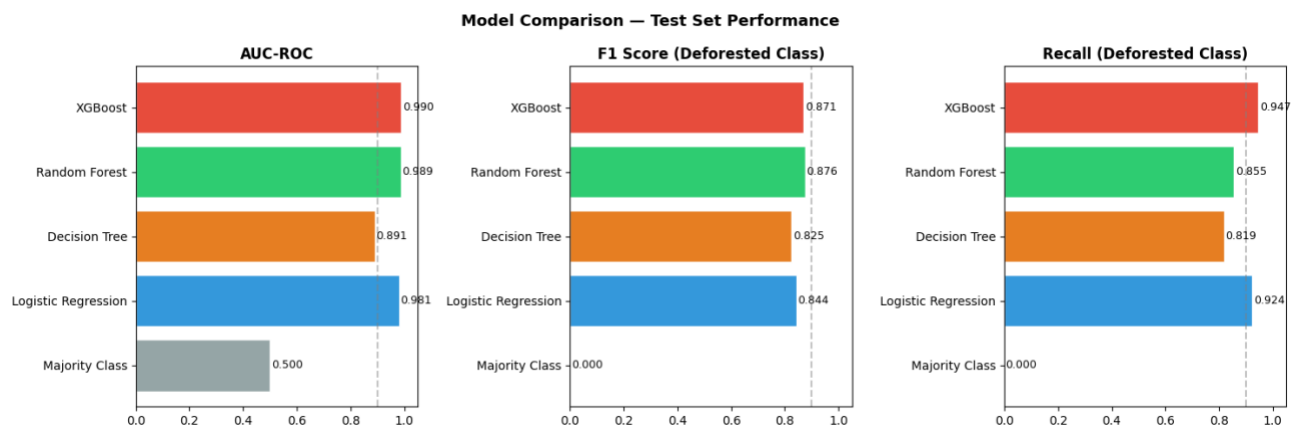


Figure 1. Distributions of scores across evaluation metrics of all models

Interpretation and Final Model Selection

XGBoost is selected as the final model. It has the highest AUC-ROC (0.990) and by far the lowest false negatives (676), meaning it misses the fewest real deforestation events of all models evaluated. In a conservation context, a false negative means a cell being actively cleared receives no enforcement or monitoring response. Minimising this error is the most important objective, making recall the decisive metric alongside AUC.

Trade-offs between models:

- Logistic Regression has surprisingly high recall (0.924) and is the most interpretable model. It casts a wide net and catches most deforested cells, but at the cost of 3,365 false alarms. A strong choice if explaining predictions to non-technical stakeholders is the priority.
- Decision Tree is the weakest real model with the lowest AUC (0.891) and high false negatives (2,285). Unlimited depth caused overfitting that hurt generalisation. Not recommended.
- Random Forest has the highest precision (0.898) and highest F1 (0.876). It is the most conservative model, flagging cells only when very confident. Fewer false alarms but misses more real deforestation events (1,833 false negatives versus XGBoost's 676).
- XGBoost balances the highest AUC with the best recall. The 2,879 false positives are an acceptable cost: monitoring a cell that turns out to be intact wastes some resources, but missing 676 real clearing events versus 1,833 is a significant conservation gain.

Section 5: Error Analysis

Error analysis investigates where and why the final model makes mistakes. Knowing that XGBoost achieves 94.7% recall is useful, but understanding what characterises the 5.3% of deforested cells it still misses is what makes the model actionable in practice. This section examines feature importance to understand what drives predictions, then compares the characteristics of missed cells against correctly caught ones to identify systematic failure patterns.

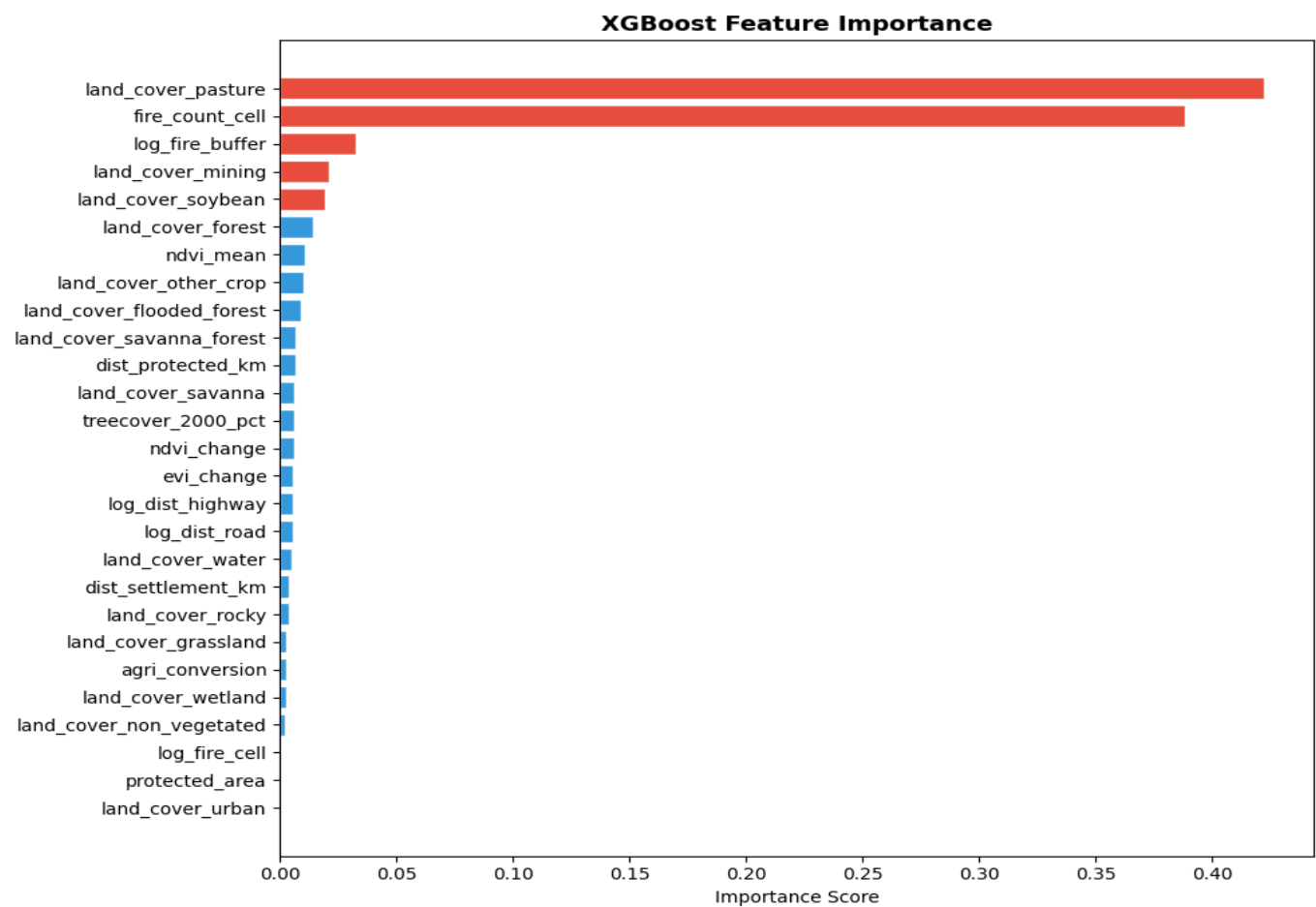


Figure 2. Features distributed respective to their importance in XGBoost model

5.1 Feature Importance

The two dominant features are land_cover_pasture (importance score 0.423) and fire_count_cell (importance score 0.388), which together account for over 81% of the model's decision-making weight. Every other feature in the model contributes less than 5% individually.

This result aligns precisely with both the EDA findings and the known ecology of deforestation along BR-163. Pasture is the primary land use conversion pathway in this corridor: forest is cleared, burned, and converted to cattle grazing, meaning a cell already classified as pasture by MapBiomass is almost certainly one that has already undergone deforestation. Fire count is the most direct satellite-observable signal of active slash-and-burn clearing, the dominant clearing method used along the BR-163 frontier.

The remaining features, including log distance to road, NDVI change, distance to settlement, and the other land cover dummies, contribute meaningfully in aggregate but no single one approaches the influence of

pasture or fire count. This concentration of importance in two features has direct consequences for where the model fails, as the next section shows.

5.2 Error Pattern Analysis

Why Does XGBoost Miss 676 Deforested Cells?

The 676 false negatives are not randomly distributed. Comparing their feature values against the 11,973 deforested cells the model correctly identified reveals a consistent and interpretable pattern:

- `log_fire_cell`: 0.448 (missed) vs 2.055 (caught). Since fire count is the second most important feature at 38.8% importance weight, cells cleared without burning are nearly invisible to the model. Mechanical clearing, selective logging, and chainsaw-based clearing leave no fire signal in the FIRMS satellite data.
- `log_dist_road`: 1.419 (missed) vs 0.898 (caught). Missed cells are systematically farther from roads. The model has correctly learned that deforestation is strongly concentrated near road access points, but this also means it underestimates risk for remote frontier encroachment.
- `log_dist_highway`: 4.391 (missed) vs 3.986 (caught). Missed cells sit deeper in the forest interior, farther from the BR-163 corridor itself. These are likely the leading edge of the deforestation frontier where clearing is earliest and least detectable.
- `agri_conversion`: 0.001 (missed) vs 0.206 (caught). Missed cells have almost no agricultural conversion flag from MapBiomas, meaning they have not yet been reclassified as farmland. This suggests they are in the very earliest stages of clearing before any agricultural activity has been formally recorded.
- `ndvi_change`: -0.010 (missed) vs -0.049 (caught). Missed cells show almost no measurable vegetation decline. This is consistent with early-stage clearing where canopy removal is partial, or with secondary regrowth that temporarily masks the vegetation loss signal.

Summary: Why Cells Are Missed

XGBoost systematically misses deforestation that is remote, fire-free, early-stage, and not yet recorded as agricultural conversion. These are precisely the hardest cases to detect from satellite data because they have not yet produced the observable signals the model was trained to recognise. This is not a failure of model tuning but a fundamental limitation of the available features.

Why Does XGBoost Produce 2,879 False Positives?

False positives are intact cells the model wrongly flags as deforested. Given that the model assigns the highest importance to pasture cover and fire activity, the most likely explanation is that these cells sit near roads, show moderate fire counts from controlled burns or natural savanna fires, and may have some pasture coverage from historical land use, but are not currently undergoing active clearing. The model's aggressive stance toward the minority class, which is what gives it the highest recall of all models, also means it flags ambiguous cells on the side of caution. In a conservation context this is the correct bias: directing monitoring resources toward a cell that turns out to be intact wastes some operational effort, but missing a real clearing event allows deforestation to go unenforced.

Section 6: Improvements and Next Steps

The current pipeline performs well as a first operational model, but several concrete improvements would meaningfully increase both its accuracy and its real-world reliability.

6.1 Additional Feature Engineering

Spatial lag features represent the single most impactful improvement available within the current modelling framework. The EDA confirmed that deforestation propagates spatially: a cell's neighbours' land cover state is highly predictive of its own future clearing risk. Currently the model treats every grid cell as completely independent with no knowledge of what its neighbours look like. Adding features such as the average deforestation rate in the surrounding 5x5 km block or the number of already-deforested neighbours would directly encode the active frontier dynamic. This improvement would most likely reduce false negatives for remote cells being gradually encroached upon, which is exactly the failure mode identified in Section 5.

Temporal fire trajectory would address the false positive problem. Currently `fire_count_cell` collapses the entire 10-year FIRMS fire record into a single cumulative count. This means a cell with stable natural savanna burning looks identical to a cell with rapidly accelerating fire activity in the last two years. Splitting fire count into an early period (2015 to 2019) and a late period (2020 to 2024) would allow the model to distinguish cells where fire is intensifying from those with historically stable burning patterns.

Distance to the nearest deforested neighbour would encode frontier proximity more precisely than highway distance alone. A cell could be far from a highway but immediately adjacent to a recently deforested block, making it extremely high risk in ways the current feature set cannot capture.

6.2 More Advanced Models

Spatial Graph Neural Networks would be the theoretically correct modelling approach for this problem. By treating the grid as a graph where edges connect neighbouring cells, a GNN could model cell-to-cell deforestation propagation explicitly rather than approximating it through engineered spatial lag features.

Ensemble stacking would combine the complementary strengths of XGBoost and Random Forest. XGBoost achieves higher recall (0.947 vs 0.855) while Random Forest achieves higher precision and F1. A stacked ensemble using both models' predicted probabilities as inputs to a meta-learner could inherit both strengths simultaneously.

Threshold optimisation offers an immediate and low-cost improvement without any retraining. Lowering the classification threshold from the default 0.5 to 0.3 or 0.35 would flag more cells as deforested, increasing recall at the cost of more false positives. Given that minimising false negatives is the primary operational objective, this trade-off is almost always worth making in a conservation deployment context.

6.3 Better Data

Socioeconomic drivers represent the most significant gap in the current feature set. Commodity prices for soy and beef, land tenure records, municipal governance quality indices, and cattle density are all well-documented drivers of Amazon deforestation that are entirely absent from this dataset. Adding a municipal-level soy price index or cattle density proxy would likely improve model performance meaningfully and would also improve interpretability for policymakers.

Spatial block cross-validation should replace the current stratified random split in any future iteration. Adjacent grid cells share similar environmental conditions by definition. Random splitting allows the model to train on cells immediately adjacent to test cells, which is a form of spatial leakage that inflates the reported AUC and recall figures. A proper geographic block holdout would produce a more honest and conservative estimate of generalisation performance.

Correcting the WDPA protected area join is a targeted fix that would restore a theoretically important predictor. The near-zero mean of the `protected_area` feature (mean = 0.00002) strongly suggests a coordinate reference system mismatch between the WDPA polygon layer and the BR-163 grid. Protected area status is one of the most important institutional predictors of deforestation and fixing the spatial join would restore this signal.

6.4 Deployment Considerations

In practice this model would run on a quarterly update cycle, ingesting the latest NASA FIRMS fire detections and MODIS NDVI composites as they become available. The output would be a georeferenced risk raster covering the entire BR-163 corridor, delivered to IBAMA or state-level environmental secretariats as a prioritisation layer. Rather than attempting to monitor all 342,819 grid cells equally, enforcement teams would focus attention on the top 5% of cells by predicted deforestation probability.

6.5 Monitoring and Model Drift

Deforestation patterns are not static. Commodity price cycles, changes in federal enforcement policy, road expansion, and political transitions all alter where and how fast clearing occurs along BR-163. The model should be retrained annually on updated Hansen GFC forest loss year data. Drift detection should monitor the distribution of predicted deforestation probabilities over time. If the proportion of cells flagged as high risk changes sharply between quarterly runs without a corresponding change in input data, this signals that the underlying pattern has shifted and the model is no longer reliably calibrated.

Section 7: Conclusion

This project built and evaluated a binary classification pipeline to predict deforestation risk across 342,819 grid cells along Brazil's BR-163 highway corridor, using features derived from satellite fire detection, road proximity, vegetation indices, and land cover history spanning 2015 to 2024.

Final Model Selection

XGBoost (learning_rate=0.1, max_depth=6, n_estimators=500) was selected as the final model based on three results that matter most for this conservation problem.

First, it achieved the highest AUC-ROC of 0.990 across all models evaluated, meaning it correctly ranks a randomly chosen deforested cell above a randomly chosen intact cell 99.0% of the time across every possible classification threshold simultaneously.

Second, it achieved the highest recall of 0.947, correctly identifying 11,973 out of 12,649 genuinely deforested cells in the held-out test set. Recall is the decisive metric here because every missed deforested cell is a real clearing event that receives no monitoring or enforcement response in an operational setting.

Third, it produced the lowest false negative count of all models at 676, compared to 963 for Logistic Regression, 1,833 for Random Forest, and 2,285 for the Decision Tree. Reducing false negatives from 1,833 to 676 means 1,157 additional deforestation events are caught and flagged for action. In a conservation context where missed deforestation allows clearing to continue undetected and unenforced, this difference is operationally significant.

Random Forest achieved a higher F1 score (0.876 vs 0.871) and fewer false positives (1,231 vs 2,879), but its false negative count of 1,833 makes it a less suitable choice for this specific problem. Minimising missed deforestation takes priority over minimising false alarms, and XGBoost delivers on that priority more effectively than any alternative evaluated.

Is Performance Sufficient for Operational Use?

The model is strong enough to serve as a prioritisation tool, directing monitoring and enforcement resources toward the highest-risk cells in the corridor rather than surveying all 342,819 cells uniformly. An AUC of 0.990 and recall of 0.947 represent genuine and reliable discrimination that would add meaningful value to conservation operations.

However, the model is not reliable enough for fully automated enforcement decisions without human validation. The 676 remaining false negatives represent real clearing events the model lets through, and the 2,879 false positives represent intact cells that would waste inspection resources if acted upon automatically. Human review of flagged cells before any enforcement action remains essential.

It is also important to note that the reported performance figures are optimistic. The current stratified random train-test split does not account for spatial autocorrelation between adjacent grid cells. A proper geographic block holdout would likely produce lower but more honest AUC and recall estimates, reflecting how the model truly generalises to genuinely unseen areas of the corridor.

Key Limitations

The model relies heavily on two features, **land_cover_pasture** (42.3% importance) and **fire_count_cell** (38.8% importance), which together account for over 81% of its decision-making. This concentration means the model is effectively blind to deforestation that does not involve fire or pasture conversion. The error analysis confirmed this directly: missed cells had a mean log fire count of 0.448 compared to 2.055 for correctly caught cells, and a mean agricultural conversion rate of 0.001 compared to 0.206. Mechanical clearing, selective logging, and early-stage frontier encroachment that has not yet produced a fire or agricultural conversion signal are the cases the model consistently fails to detect.

The `protected_area` feature is effectively non-functional due to a suspected coordinate reference system mismatch in the WDPA spatial join, removing a theoretically important institutional predictor. Socioeconomic drivers including commodity prices, land tenure records, and cattle density are absent from the dataset despite being well-documented deforestation drivers in the BR-163 literature.

Most Important Next Step

The single most impactful next step is to add **spatial lag features** encoding neighbourhood deforestation rates and to implement spatial block cross-validation. These two changes address the model's primary failure mode and its most significant evaluation bias simultaneously. Spatial lag features would give the model awareness of the active deforestation frontier, directly reducing false negatives for the remote, early-stage clearing events it currently misses. Spatial block cross-validation would replace the current optimistic performance estimate with one that honestly reflects generalisation to unseen geographic areas, which is the metric that actually matters for operational deployment along BR-163.

Critical Note on False Negatives

Throughout this project, **minimising false negatives** has been treated as the primary operational concern above all other metrics. A false negative means a deforested cell goes undetected and receives no enforcement response while clearing continues. With 676 false negatives in the test set, XGBoost is the strongest model for this objective. Any future iteration of this pipeline should continue to treat recall and false negative count as the primary measures of operational success.